Chapter 5

QBIO 305 FA26

Summary of lecture slides

Reminder: Simple random sample

- Definition
  - Every member of the population has the same chance of being included (sample is representative of the population)
  - Members of the sample are chosen independently of each other
- If we take a simple random sample of a random value distribution, another way to say this is the elements of the sample are independent and identically distributed (i.i.d.)
  - Identically distributed means they're from the same distribution

Law of Large Numbers

- Let X_1, X_2, …, X_n be independent and identically distributed random variables, each with mean μ. As the sample size n increases to infinity

$$\overline{X_n} = \frac{X_1 + X_2 + \ \ldots + X_n}{n} = \sum_{i=1}^{n} \frac{X_i}{n} \ \rightarrow \mu$$

- The sample mean $(\overline{X_n})$ converges to the random variable mean (μ)
  - In other words, for a large enough sample, the sample mean gets close to the random variable mean

LLN: Bernoulli example

- Suppose X_1, …, X_n are independent Bernoulli random variables
  - X_i = 1 with probability p (success, heads, etc.)
  - X_i = 0 with probability 1-p (failture, tails, etc.)
- Recall that the random variable mean = EX_i = 1*p + 0*(1-p) = p
- LLN says as n → infinity, the sample mean converges to the random variable mean:

$$\frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} \rightarrow p$$

LLN: 6-sided die example

- Suppose X_1, …, X_n are the numbers when we independently roll a fair die 6-sided die
- Random variable mean = EX_i = 1/6+2/6+3/6+4/6+5/6+6/6 = 3.5
- LLN says as n → infinity, the sample mean converges to the random variable mean:

$$\frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} \rightarrow 3.5$$

Central Limit Theorem

- X_1, …, X_n are independently and identically distributed random variables, each with mean μ and standard deviation σ. As sample size n increases to infinity:

$$\sum_{i=1}^{n} \frac{(X_i - \mu)}{\sqrt{n}\sigma} \Rightarrow Normal(0,1)$$

where Normal(0,1) is a normal random variable with mean 0 and standard deviation 1 (shortened N(0,1)).

Convergence

- What does it mean for one random variable to converge to another?
- The distribution functions of the two random variables converge

Normal distribution

- For X = $\mathrm{N}(\mu, \sigma)$ and constants a and b:
  - aX+b is also a normal random variable with mean aμ+b and standard deviation aσ: $\mathrm{N}(\mathrm{a}\mu + b, a\sigma)$

For large sample sizes n

$$\sum_{i=1}^{n} \frac{(X_i - \mu)}{\sqrt{n}\sigma} \approx N(0,1)$$

$$\sum_{i=1}^{n} (X_i - \mu) \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \mu \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i - n\mu \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i \approx N(n\mu, \sqrt{n}\sigma)$$

The **sum** of independent random variables is approximately **normally distributed** with mean $n\mu$ and standard deviation $\sqrt{n}\sigma$

Normal distribution

- Ubiquitous
- Sum of many independent variables results in a normal random variable
- E.g. sum of cells for height, measure of blood pressure, measure of cholesterol, batting averages

For large sample sizes *n*,

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}}\, N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The sample mean is also **normally distributed**, now the mean is μ and the standard deviation is $\sigma/\sqrt{n}$

For large sample sizes n,

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}} N(0,1)$$

The Central Limit Theorem tells us **both** about the sum of independent random variables and their mean

The only difference is whether we divide by the sample size n

For large sample sizes n,

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}}\, N(0,1)$$

The Law of Large Numbers tells us the sample mean is close to μ

The Central Limit Theorem tells us **how close**: $\frac{\sigma}{\sqrt{n}}\, N(0,1)$

Possible problems when applying the Central Limit Theorem

- Is an additive model appropriate?
- Are the random variables identically distributed? (baseball example)
- Are the random variables independent? (baseball example)
- How big does the sample size n have to be for the approximation to be valid?